

Gihyeok (Peter) Lee

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EDUCATION

University of Waterloo

Honours Nanotechnology Engineering, Co-operative Program, BAsC.

Waterloo, ON

Sept. 2023 – Apr. 2028

EXPERIENCE

SLS Materials Intern

Formlabs

Sept. 2025 – Apr. 2026

Somerville, MA

- Led material optimization of a high-performance polyamide powder, increasing elongation at break from <2% to 8% through base-polymer and additive reformulation while maintaining SLS processability
- Led 16× scale-up of powder-blending processes from benchtop to industrial mixers for next-generation SLS powders, ensuring formulation consistency and enabling production-scale manufacturing
- Contributed to the launch of a high-ductility nylon powder by identifying process conditions that increased defect-free part yield by 25% and improved print and post-processing performance
- Investigated the effect of a powder-processing additive on elevated-temperature cohesion, identifying conditions that expanded the usable SLS processing window while maintaining mechanical performance
- Screened candidate polymer systems through mechanical and processability testing, identifying low-value development paths and supporting material down-selection for further R&D

Process Core Member

HackerFab

May 2026 – Present

Waterloo, ON

- Developed and standardized a photoresist spin-coating process on diced silicon by defining dispense, spin, bake, and handling parameters; reduced patterning failures by 70% and documented the final process in a team SOP
- Fabricating MOS capacitors using thermal oxidation, photolithography, wet etching, and aluminum sputtering, with planned device characterization through C–V and I–V measurements
- Building a low-cost laser-scanning microscope using a DVD optical pickup and piezoelectric sample stage; evaluating photodiode readout, stage actuation, and control methods

Undergraduate Research Assistant

Maikawa Lab, University of Toronto

Feb. 2025 – Apr. 2025

Toronto, ON

- Designed and optimized a hydrogel particle-processing system using 3D-printed components, achieving a 70% reduction in particle size and 90% reduction in processing time
- Developed and tested seven iterative grinder-head designs using SolidWorks, optimizing for reduced particle size and polydispersity; validated results using microscopy and ImageJ analysis
- Performed quantitative NMR experiments to evaluate UV-curing conditions and correlate polymerization behavior with photoinitiator exposure time

PROJECTS

Automated Particle Detection and Measurement System | *Python, OpenCV, NumPy, pandas*

Apr. 2025

- Developed a Python-based microscopy analysis tool using OpenCV to automatically detect and measure hydrogel particles
- Implemented Otsu thresholding and contour detection to segment particle boundaries and quantify particle dimensions
- Automated particle measurement and Excel data export to reduce repetitive manual analysis and accelerate comparison of experimental conditions

TECHNICAL SKILLS

Programming: Python, MATLAB, Java, OpenCV, NumPy, pandas

Engineering Software: SolidWorks, Onshape, ImageJ, TRIOS

Materials & Characterization: SEM, DSC, TGA, qNMR, powder rheology, optical microscopy, tensile testing

Fabrication: Photolithography, spin coating, thermal oxidation, wet etching, sputter deposition, tube furnace